Thomas Emerick

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EDUCATION

Louisiana State University, Baton Rouge, LA **Expected May 2027**
B.S. in Mathematics - Data Science Concentration GPA: 3.96/4.0

- 3x Louisiana-Mississippi Section Winner - Putnam Mathematical Competition
- Ogden Honors College | Tom and Peg Madden Scholarship for Mathematical Research
- Resident Assistant for 3 years in the Laville Honors House
- Relevant Coursework: Probability, Mathematical Statistics, Numerical Analysis, Linear Algebra, Optimization Theory, Abstract Algebra I-II, Discrete Math

WORK EXPERIENCE

Kelly Services Troy, MI
AI Engineering and Business Analyst Intern Jun 2026 – PRESENT

- Redesigned and validated 20+ ServiceNow notifications, standardizing communications for systems supporting 800+ developers and 4,800+ employees worldwide
- Built a six-tool Python/FastMCP server that converts natural-language requirements into validated, previewed, approval-gated, and inactive-by-default ServiceNow notification records

LSU Football Baton Rouge, LA
Performance Innovation and Data Analytics Jan 2025 – PRESENT

- Developed and evaluated regression models in Python and JMP to predict NFL Draft selection and quantify the player attributes most associated with professional and collegiate success
- Analyzed transfer portal outcomes and built Tableau dashboards to support recruiting and retention efforts
- Automated the retrieval, normalization, and integration of confidential performance and player-tracking data, saving 100+ hours of manual preparation per season through a single-command pipeline

RESEARCH EXPERIENCE

Exactification of Projective Line Packings May 2026

- Added 7 exact closed-form solutions to the Game of Sloanes and Subspace Packings Databases, replacing numerically verified entries with algebraically certified ones
- Adapted Mathematica and MATLAB scripts to build a pipeline converting numerically-approximate line-packing solutions into exact, symbolically verified closed-form results
- Presented findings at the 10th Iberian Mathematical Meeting in Gijón, Spain

Exchange-Traded Fund Portfolio Optimization May 2026

- Built a Python and CVXPY portfolio-optimization pipeline to allocate capital across exchange-traded funds under return, volatility, diversification, and position-size constraints
- Implemented historical backtesting and periodic rebalancing to compare optimized portfolios against benchmark and equal-weight strategies

Expected Value Modeling of Batted-Ball Outcomes Dec 2025

- Modeled expected batting average (xBA) and expected weighted on-base average (xwOBA) from exit velocity and launch angle using k-nearest neighbors classification and generalized additive models
- Built interpretable probability surfaces from batted-ball data to support player evaluation and development

SKILLS

Data & Analysis: Python, SQL (DuckDB), pandas, numpy, scikit-learn, Excel, R
Data Engineering: API ingestion, ETL pipelines, data cleaning & normalization
Visualization: Tableau, Power BI, Streamlit